MILITARY SPECIFICATION

PROPELLANT CHARGE, NUCLEAR COMPONENT, TRAINING

This specficiation is mandatory for use by the activities of the Army Munitions Command.

1. SCOPE

1.1 Scope. This specification covers special requirements, quality assurance provisions, and the preparation for delivery criteria for one type of propellant charge used on the M423 projectile.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation Methods of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition

DRAWINGS

ORDNANCE CORPS

8786584 - Spacer
8787136 - Propellant Case
8787137 - Liner, Case
8787138 - Disc, Inner
8787139 - Disc, Front

FSC 1110

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:40Z
Check the source to verify that this is the current version before use.

8787208 - Propellant Charge, Nuclear Component, Training
8811088 - Carton, Packing, Ammunition, for Propellant Charge, Nuclear Component Training
ET - 8787208 - Equipment List

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The propellant charge shall comply with all requirements specified on Drawing (dwg 8787208), with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Tool manufacturing control (T.M.C.). When T.M.C. is specified herein, five consecutive parts which comply with drawing dimensions will constitute proper initial tool control acceptance. During normal production, five parts shall be selected at the start and at the end of each day's production (for a minimum of 10 parts) and examined as specified in the applicable paragraph in 4.3.2.1. In addition, five consecutive parts shall be examined as specified in the applicable paragraph in 4.3.2.1 whenever the manufacturing tool requires maintenance. Failure of any part to comply with any of the examinations specified in 4.3.2.1 shall reject the lot of production under consideration.

3.4 Workmanship.

3.4.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.4.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.4.3 Burrs. Parts shall be free from burrs.

3.4.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.4.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:40Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not Applicable.

4.3 Product acceptance inspection .

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table I
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:40Z
Check the source to verify that this is the current version before use.

Table II

Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Propellant case (see dwg. 8787136) covering a detail of dwg. 8787208.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major
101. Diameter of right cavity.....Gage
102. Depth of right cavity, min.....Gage
103. Material (see 3.1).....Visual

Minor:
201. Marking.....Visual
202. Workmanship (see 3.4).....Visual/Manual

4.3.2.1.2 Propellant charge, nuclear component, training (see dwg. 8787208).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Diameter of double seam, max.....Gage
102. Length, max.....Gage
103. Propellant weight.....Gage
104. Objective evidence (see 6.2) for protective finish in accordance with dwg. 8787208.....Visual
105. Material (see 3.1).....Visual

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:40Z
Check the source to verify that this is the current version before use.

Minor:
201. Marking.....Visual
202. Workmanship (see 3.4).....Visual/Manual

4.3.2.1.3 Carton, packing, ammunition, for propellant charge, nuclear component, training (see dwg. 8811088)

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Carton properly sealed.....Visual
102. Contents tight in carton.....Manual
103. Croners blunted on carton.....Visual/Manual
104. Proper heat seal on bag.....Visual
105. Air exhausted from bag.....Manual
106. Objective evidence (see 6.2) for heat seal test in accordance with MIL-P-116.....Visual
107. Packaging materials (see 3.1).....Visual

Minor:
201. Workmanship (see 3.4).....Visual/Manual

4.3.3 Testing, Not Applicable.
4.3.4 Acceptance inspection equipment.
4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET 8787208. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government Representative authorized by the procuring agency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:40Z
Check the source to verify that this is the current version before use.

4.4 Test methods. Not applicable.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The propellant charge shall be packaged in accordance with dwg. 8811088 using the specified moisture barrier.

5.1.2 Level C. The propellant charge shall be packaged in accordance with dwg. 8811088 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The propellant charge shall be packed in accordance with dwg. 8811088.

5.3 Marking. Marking shall be as specified in accordance with dwg. 8811088.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specifications.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2.).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.4 Supersession data. This specification includes the requirements of Purchase Description PA-PD-1128 (Rev 2), dated 21 November 1960.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1110-A019

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:40Z
Check the source to verify that this is the current version before use.